

METABOLISM

Module map

- Module 06: Metabolism
- Connected lab: lab-06_metabolism.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 06

Metabolism

1

Energy and
thermodynamics

2

ATP as
cellular
coupling

3

Enzymes
and
activation
energy

4

Regulation
by
conditions
and
inhibitors

5

Photosynthesis
and
respiration as
linked
pathways

Lab evidence

lab-06_metabolism.md

METABOLISM

Learning objectives

- Apply thermodynamic ideas to living cells.
- Explain ATP as an energy-coupling molecule.
- Describe how enzymes change reaction rates.
- Predict how environmental conditions affect enzyme activity.
- Connect photosynthesis and cellular respiration in matter and energy flow.

OBJECTIVE 1

Apply thermodynamic ideas to living cells.

OBJECTIVE 2

Explain ATP as an energy-coupling molecule.

OBJECTIVE 3

Describe how enzymes change reaction rates.

OBJECTIVE 4

Predict how environmental conditions affect enzyme activity.

OBJECTIVE 5

Connect photosynthesis and cellular respiration in matter and energy flow.

METABOLISM

Topic sequence

- Energy and thermodynamics: Cells transform energy while obeying thermodynamic constraints.
- ATP as cellular coupling: ATP couples energy-releasing reactions to energy-requiring cellular work.
- Enzymes and activation energy: Enzymes lower activation energy without being consumed.
- Regulation by conditions and inhibitors: Temperature, pH, substrate concentration, and inhibitors alter enzyme function.
- Photosynthesis and respiration as linked pathways: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

01

Energy and thermodynamics

Cells transform energy while obeying thermodynamic constraints.

02

ATP as cellular coupling

ATP couples energy-releasing reactions to energy-requiring cellular work.

03

Enzymes and activation energy

Enzymes lower activation energy without being consumed.

04

Regulation by conditions and inhibitors

Temperature, pH, substrate concentration, and inhibitors alter enzyme function.

05

Photosynthesis and respiration as linked pathways

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

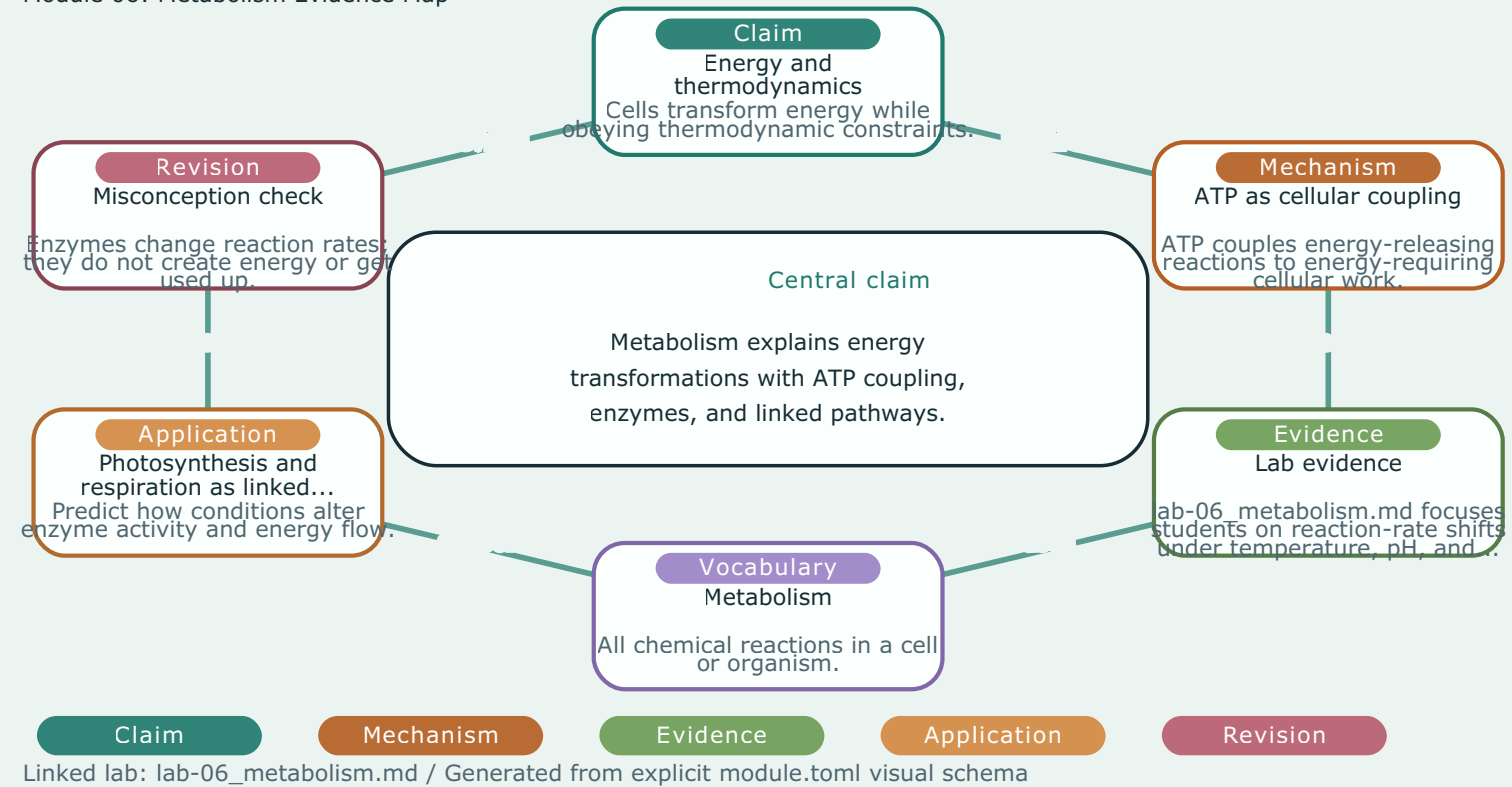
METABOLISM

Module 06: Metabolism Evidence Map

- Module focus: Metabolism
- Central claim: Metabolism explains energy transformations with ATP coupling, enzymes, and linked pathways.
- Key nodes to connect: Energy and thermodynamics, ATP as cellular coupling, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-06_metabolism.md supplies an evidence surface for the map.

Module 06 / Concept Map

Module 06: Metabolism Evidence Map



Teaching note: Ask students to explain one edge in their own words before moving on.

METABOLISM

Module 06: Metabolism Reasoning Sequence

- Module focus: Metabolism
- Trace the model: Energy and thermodynamics -> ATP as cellular coupling -> Enzymes and activation energy -> Regulation by conditions and inhibitors
- Inputs become outputs: Energy and thermodynamics, ATP as cellular coupling -> Apply thermodynamic ideas to living cells., Explain ATP as an energy-coupling molecule.
- Feedback check: lab-06_metabolism.md evidence checks whether students can explain enzymes and activation energy.
- Lab connection: lab-06_metabolism.md gives students a process to observe or test.

Module 06 / Process Model

Module 06: Metabolism Reasoning Sequence

Inputs

- Energy and thermodynamics
- ATP as cellular coupling
- Evidence from lab-06_metabolism.md

Outputs

- Apply thermodynamic ideas to living cells.
- Explain ATP as an energy-coupling molecule.
- Connect photosynthesis and cellular respiration in matter and...

Reasoning sequence

1

Energy and thermodynamics

Cells transform energy, obeying thermodynamic principles.

2

ATP as cellular coupling

ATP couples energy-related reactions to energy-requiring cellular work.

3

Enzymes and activation energy

Enzymes lower activation energy without being consumed.

4

Regulation by conditions and inhibitors

Temperature, pH, substrate concentration, and inhibitors alter enzyme function.

5

Photosynthesis and respiration as linked.

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

Feedback

lab-06_metabolism.md evidence checks whether students can explain enzymes and activation energy.; Retrieval questions reveal whether terms...

Constraint

Enzymes change reaction rates; they do not create energy or get used up.; Photosynthesis builds sugar while respiration breaks fuel down to...

Linked lab: lab-06_metabolism.md

METABOLISM

Terms as evidence handles

- Metabolism: All chemical reactions in a cell or organism.
- ATP: Cellular energy-coupling molecule.
- Activation energy: Energy required to start a reaction.
- Enzyme: Protein or RNA catalyst that speeds a reaction.
- Substrate: Reactant that binds an enzyme active site.
- Denaturation: Loss of shape that disrupts function.

METABOLISM

All chemical reactions in a cell or organism.

ATP

Cellular energy-coupling molecule.

ACTIVATION ENERGY

Energy required to start a reaction.

ENZYME

Protein or RNA catalyst that speeds a reaction.

SUBSTRATE

Reactant that binds an enzyme active site.

DENATURATION

Loss of shape that disrupts function.

METABOLISM

Lab connection

- Lab file: lab-06_metabolism.md
- Lab evidence should test or illustrate:
Enzymes and activation energy
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Energy and thermodynamics

EVIDENCE

lab-06_metabolism.md

CLAIM

Apply thermodynamic ideas to living cells.

METABOLISM

Contrast check

- Do not stop at: Cells transform energy while obeying thermodynamic constraints.
- Stronger explanation: Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Cells transform energy while obeying thermodynamic constraints.

DEEPER

Photosynthesis builds sugar while respiration breaks fuel down to harvest usable energy.

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Module 06: Metabolism Retrieval and Lab Check

- Module focus: Metabolism
- Retrieval prompt: How do the laws of thermodynamics constrain cells?
- Answer check: Apply thermodynamic ideas to living cells.
- Required terms: Metabolism, ATP, Activation energy
- Lab connection: lab-06_metabolism.md; lab-06_metabolism.md supplies evidence for reaction-rate shifts under temperature, pH, and substrate changes.

Module 06 / Retrieval Card

Module 06: Metabolism Retrieval and Lab

Check

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

How do the laws of thermodynamics constrain cells?

Check: Apply thermodynamic ideas to living cells.

Why is ATP useful for cellular work?

Check: Explain ATP as an energy-coupling molecule.

How do enzymes lower activation energy?

Check: Describe how enzymes change reaction rates

Why does enzyme shape matter?

Check: Predict how environmental conditions affect enzyme activity

Terms to use

Metabolism, ATP, Activation energy, Enzyme, Substrate

Lab connection

lab-06_metabolism.md supplies evidence for reaction-rate shifts under temperature, pH, and substrate changes.

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

METABOLISM

Practice quiz bridge

- 1. What is the main role of ATP in a cell?
- 2. How does an enzyme speed up a chemical reaction?
- 3. Raising temperature far above normal can stop an enzyme from working because:
- 4. How are photosynthesis and cellular respiration related?

QUESTION 1**Key: B**

What is the main role of ATP in a cell?

QUESTION 2**Key: A**

How does an enzyme speed up a chemical reaction?

QUESTION 3**Key: D**

Raising temperature far above normal can stop an enzyme from working because:

QUESTION 4**Key: C**

How are photosynthesis and cellular respiration related?

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Synthesis and exit ticket

- Exit claim: explain energy and thermodynamics using evidence from lab-06_metabolism.md.
- Must include: Metabolism, ATP, Activation energy.
- Revision prompt: what evidence would change your explanation?

CLAIM

Energy and thermodynamics

EVIDENCE

lab-06_metabolism.md

REVISION

What would change your mind?